

Dataset Description

VinDr-CXR (VinBigData Chest X-ray) Dataset

1. Introduction

The **VinDr-CXR dataset** was developed to support automated detection and localization of thoracic abnormalities in chest radiographs. It was released by **VinBigData** in partnership with Hospital 108 and Hanoi Medical University Hospital, Vietnam.

What distinguishes this dataset is the availability of **radiologist-level annotations** for both:

- **Image-level classification** (disease presence/absence)
- **Object-level localization** (bounding box annotations)

By incorporating multiple expert annotations per image, the dataset addresses common challenges in medical imaging such as **label noise**, **inter-observer variability**, and annotation inconsistency.

2. Dataset Structure and Composition

The dataset contains **18,000 postero-anterior (PA) chest X-ray images** stored in DICOM format.

Split	Number of Images	Annotation Type
Training Set	15,000	Publicly available labels
Test Set	3,000	Labels withheld (benchmark evaluation)

Image Specifications

- **Format:** DICOM (.dcm / .dicom)
 - **Bit Depth:** Typically 12-bit or 16-bit
 - **Dynamic Range:** High (requires normalization for deep learning pipelines)
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3. Annotation Methodology

Training Set (15,000 Images)

- Each image was **independently reviewed by three experienced radiologists**.
- Multiple annotations per image are possible.
- Disagreements may exist between annotators (e.g., one radiologist marks a lesion while another does not).

Modeling Implication:

This multi-annotator setup introduces variability and requires robust label handling techniques such as:

- Majority voting
- Weighted Box Fusion (WBF)
- Soft-label training strategies

Test Set (3,000 Images)

- Annotated using **consensus labeling from five radiologists**.
 - Designed to serve as a high-quality **gold standard** benchmark.
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4. Target Classes and Distribution

The dataset includes **14 clinically significant thoracic findings**, along with a “No finding” category.

Below is a summarized overview of class distribution in the training data (67,914 total annotations):

Class ID Abnormality		Approx. Count
0	Aortic enlargement	7,162
1	Atelectasis	279
2	Calcification	960
3	Cardiomegaly	5,427
4	Consolidation	556
5	Interstitial Lung Disease (ILD)	1,000

Class ID	Abnormality	Approx. Count
6	Infiltration	1,247
7	Lung Opacity	2,483
8	Nodule/Mass	2,580
9	Other lesion	2,203
10	Pleural effusion	2,476
11	Pleural thickening	4,842
12	Pneumothorax	226
13	Pulmonary fibrosis	4,655
14	No finding	31,818

Distribution Characteristics

- **Most frequent class:** “No finding” (~47% of annotations)
- **Rare classes:** Pneumothorax, Atelectasis, Consolidation
- **Challenge:** Severe class imbalance

To mitigate imbalance, recommended strategies include:

- Class-weighted loss functions
- Oversampling of minority classes
- Focal loss
- Data augmentation

5. File Organization and Metadata

Annotation File (train.csv)

The train.csv file contains bounding box annotations with the following fields:

- `image_id` – Unique identifier of each X-ray image
- `class_name` – Name of abnormality
- `class_id` – Numerical class label (0–14)

- rad_id – Radiologist identifier (e.g., R8, R10)
- x_min, y_min, x_max, y_max – Bounding box coordinates

Note: For images labeled as **“No finding”**, bounding box fields are empty (NaN).

Directory Layout

/vinbigdata-chest-xray-abnormalities-detection/

```

├── train.csv
├── sample_submission.csv
├── train/
│   ├── [image_id].dicom
│   └── ...
└── test/
    ├── [image_id].dicom
    └── ...

```

6. Data Preprocessing Requirements

Before training deep learning models, the dataset requires careful preprocessing:

1. DICOM Processing

- Extract pixel arrays from DICOM files
- Handle metadata correctly
- Adjust photometric interpretation:
 - MONOCHROME1 images require inversion
 - MONOCHROME2 images are standard

2. Pixel Normalization

Since DICOM images contain high dynamic pixel values (>2000), they must be:

- Min-max normalized
- Converted to 8-bit format (0–255)

- Saved as PNG or JPEG for compatibility with standard CNN architectures

3. Image Resizing

Original resolutions often exceed 2000×2000 pixels.

For efficient model training, images are typically resized to:

- 512×512
- 768×768
- 1024×1024

depending on model architecture (e.g., ResNet, EfficientNet, YOLO-based detectors).

Summary

The VinDr-CXR dataset provides a high-quality benchmark for automated chest X-ray abnormality detection. Its strengths include:

- Multi-radiologist annotations
- Localization and classification labels
- Large-scale dataset size
- Clinically relevant findings

However, challenges such as annotation variability and class imbalance must be addressed through careful preprocessing and modeling strategies.
